AI Strategy Proposal · V2

# OEE_SCI V2
# AI Integration Strategy

Introducing artificial intelligence into the garment factory OEE monitoring system to enable predictive maintenance, anomaly detection, and AI-powered quality analysis.

| DATE | TARGET VERSION | STATUS |
| --- | --- | --- |
| 2026-03-06 | OEE_SCI_V2 | Design Phase |

# 01 Current System Architecture (V1 Analysis)

### Target Equipment

Real-time OEE data reception from computer pattern sewing machines at SCI Indonesia.

### Real-time Communication

SSE-based 5-second polling. Equipment sends data via REST API to MySQL in real time.

### OEE Calculation

Automatic calculation of Availability x Performance x Quality. Aggregated by time slot and shift with full history.

### Alert System

3 event types: Andon, Downtime, Defective. Occurrence/completion handling with real-time dashboard display.

| ITEM | DETAIL |
|---|---|
| Technology Stack | PHP + MySQL + jQuery + Chart.js + SSE |
| UI Framework | SAP Fiori Horizon Light |
| Core Data | data_oee · data_oee_rows_hourly · Downtime · Defective · Andon |
| Hierarchy | Factory → Line → Machine |
| Current AI | None (V1 Dashboard name only says "AI OEE") |

## 02 Core AI Strategy

The V1 `dashboard.php` is named "AI OEE CS DASHBOARD" but contains no actual AI functionality. In V2, a **dedicated AI Dashboard page** will be newly implemented, with AI analysis added to existing pages in phases.

### Predictive

Forecast OEE and Downtime using historical data patterns. Enables proactive response before issues occur.

Phase 1

### Detection

Real-time anomaly detection using Z-Score. Instant alert when metrics deviate from normal range.

Phase 1

### Generative

Automated AI report generation. Auto-create executive summaries and improvement recommendations.

Phase 1

## 03 New AI Dashboard Design

A new `ai_dashboard.php` will be implemented separately from the existing `dashboard.php`.

AI OEE Intelligence Dashboard — ai_dashboard.php

All Factory | All Line | All Machine | Today | All Shift | Refresh

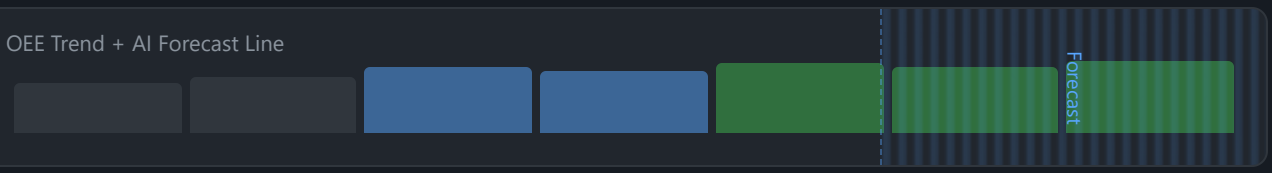
OEE Forecast (Next 4H)
**74%**
Current 76% · Decline expected

At-Risk Machines
**3**
Immediate inspection needed

Anomaly Alerts
**0**
Normal operation

AI Insights
**2**
New recommendations

OEE Trend + AI Forecast Line




## AI HEALTH INDEX BY LINE

| | | |
|---|---|---|
| Line A | 82% | Normal |
| Line B | 61% | Critical |
| Line C | 74% | Warning |
| Line D | 80% | Normal |

## MAINTENANCE PRIORITY LIST

| | | | |
|---|---|---|---|
| 1 | M-012 | Risk 89% | ~2.1H later |
| 2 | M-007 | Risk 72% | ~5.3H later |
| 3 | M-023 | Risk 65% | ~7.8H later |

# 04 Detailed AI Feature Design

## 1 OEE Prediction (Predictive OEE)

Phase 1

**Applied to**: AI Dashboard, OEE monitoring chart with forecast line added

- Learns hourly OEE patterns from the past 30 days of data
- Forecasts OEE for the next 4~8 hours using a **day-of-week x hour-of-day** matrix
- Visualizes dashed forecast interval (Confidence Interval) on the chart

**Example**: "Current OEE at 2 PM is 72% — forecast shows a decline to 68% over the next 2 hours (based on Monday afternoon pattern)"

## 2 Predictive Maintenance

Phase 1

**Applied to**: AI Dashboard at-risk machine panel, Downtime monitoring with risk indicator column

- Analyzes **Downtime occurrence cycle patterns** per machine
- Compares elapsed time since last Downtime against average cycle
- Calculates **risk score (0~100)** in real time

| SCORE | LEVEL | DISPLAY |
|---|---|---|
| 80~100 | Critical | Red |
| 50~79 | Warning | Yellow |
| 0~49 | Normal | Green |

### 3 Anomaly Detection

Phase 1

**Applied to**: AI Dashboard real-time alert panel, existing monitoring pages

- **Z-Score based**: Instant detection when OEE, Defective rate, or Downtime frequency exceeds ±2σ
- **Pattern anomaly**: Detects a drop of -15% or more in OEE compared to the previous hour
- **Cascading anomaly**: Detects multiple machines failing simultaneously on the same line within 30 minutes

### 4 Quality Sentinel

Phase 1

**Applied to**: Defective monitoring page with an AI analysis tab added

- Defective occurrence pattern clustering (time slot x machine x defective type)
- **Automated Pareto analysis**: "80% of defectives originate from these 3 types"
- Correlation analysis: predicts OEE impact when a specific Defective type increases

### 5 AI Report Generator

Phase 1

**Applied to**: Report page with an "AI Analysis Report" button added

- Automatically analyzes OEE, Downtime, and Defective data for the selected period
- Auto-generates an **executive summary report**
- AI commentary sheet automatically included when exporting to Excel

## 05 Implementation Roadmap

### Phase 1 Statistics-based AI

Utilizes existing DB data — fast to implement

- OEE Prediction (Moving Avg.) ●●○
- Anomaly Detection (Z-Score) ●●○
- Maintenance Risk Score ●●○
- AI Health Index UI ●○○
- Quality Sentinel ●●○
- AI Report Generator ●●○

### Phase 2 Advanced AI

Real-time AI analysis and optimization recommendations

- Real-time Streaming AI Analysis ●●●
- Production Optimization Suggestions ●●●

● Medium ● Very High